| DS | **Session** | Sorting | **Question Information** | Level 1 Challenge 19 |
|---|---|---|---|---|

Problem Description:

**Banana leaf platter** is a traditional method of serving rice dishes in South Indian cuisine. Due to the migration of South Indians, banana leaf rice can also be found in areas with significant ethnic South Indian diaspora such as Malaysia and Singapore.

Irfan is a banana leaf sales person.
he has N stacks of banana leafs.

Each stack contains K leafs.

Each leaf has a positive beauty value, describing how attractive it looks.

Irfan would like to take exactly P leafs to use for lunch today. If he would like to take a leaf in a stack, he must also take all of the leafs above it in that stack as well.

Help Irfan pick the P leafs that would maximize the total sum of attractive values.

Constraints:
$1 \leq T \leq 100$.
$1 \leq K \leq 30$.
$1 \leq P \leq N * K$.
$1 \leq N \leq 50$.

Input Format:
The first line of the input gives the number of test cases, T. T test cases follow. Each test case begins with a line containing the three integers N, K and

P. Then, N lines follow. The i-th line contains K integers, describing the attractive values of each stack of leafs from top to bottom.

Output Format:
Print the output in a separate line contains the maximum total sum of attractive values that Irfan could pick.

## Logical Test Cases

### Test Case 1

INPUT (STDIN)

```
2
2 4 5
10 10 100 30
80 50 10 50
3 2 3
80 80
20 20
15 50
```

EXPECTED OUTPUT

```
250
180
```

### Test Case 2

INPUT (STDIN)

```
2
2 4 5
20 20 70 30
60 30 10 50
3 2 3
60 70
15 40
20 10
```

EXPECTED OUTPUT

```
200
150
```

## Mandatory Test Cases

### Test Case 1

KEYWORD

```
int max(int a,int b)
```

### Test Case 2

KEYWORD

```
for(int i = 0;i < n;i++)
```

Complexity Test Cases

| Test Case 1 | Test Case 2 | Test Case 3 |
| --- | --- | --- |
| CYCLOMATIC COMPLEXITY | TOKEN COUNT | NLOC |
| 2 | 380 | 47 |

## Code

solution.c

```c
#include <stdio.h>

int max(int a, int b) { return (a > b) ? a : b; }

int main() {
    int T;
    if (scanf("%d", &T) != 1) return 0;
    while (T--) {
        int n, k, p;
        scanf("%d %d %d", &n, &k, &p);
        int dp[1005] = {0};
        for (int i = 0; i < n; i++) {
            int pref[35] = {0};
            for (int j = 1; j <= k; j++) {
                int val; scanf("%d", &val);
                pref[j] = pref[j - 1] + val;
            }
            for (int curP = p; curP >= 0; curP--) {
                for (int take = 1; take <= k && take <= curP; take++) {
                    dp[curP] = max(dp[curP], dp[curP - take] + pref[take]);
                }
            }
        }
        printf("%d\n", dp[p]);
    }
    return 0;
}
```

## Output

```
Problems  Output  Debug Console  Terminal  Ports        powershell
PS C:\C Programming> gcc solution.c -o solution
PS C:\C Programming> .\solution
2
2 4 5
10 10 100 30
80 50 10 50
3 2 3
80 80
20 20
20 10
250
180
PS C:\C Programming>
```